TRAIN100 Results

Every model trained to (nearly) 100 % training accuracy: accuracy, F1, recall, confusion matrices and model structure

Four-class LVEF classification from 12-lead ECG printouts. Development data: 837 ECGs, patient-grouped 5-fold cross-validation × 3 repeats.

The locked test set was not used. EchoNext was used only on the owner's laptop (Section 9).

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1 What was done

Every model of the project that could be retrained was trained again, with no early stopping, no weight decay, no augmentation and as many epochs or iterations as it needed, until its training accuracy reached (nearly) 100 %. It was then scored on the same held-out folds as the original early-stopped model. The rules were declared before any run (`AMENDMENT_TRAIN100.md`, branch `research/train100`).

Metrics. *Held-out accuracy*: share of held-out ECGs classified correctly (four classes). *Macro-F1*: mean of the four per-class F1 scores (F1 = harmonic mean of precision and recall), so every class counts equally. *Mild F1*: F1 of the hardest class. *Balanced accuracy*: mean of the four recalls. *Recall*: share of each true class found.

In one sentence. Every flexible model now fits its training data perfectly, but none becomes reliably better on new patients: the best four-class accuracy is LGBM-FEAT with 0.548, the best macro-F1

FUSION-MLP with 0.438 and the best Mild F1 ECGF-linear with 0.282, all at or below the early-stopped originals.

2 Main models ranked by held-out accuracy

Table 1: Main models trained to (nearly) 100 % training accuracy, ranked by held-out accuracy (development cross-validation, 5 folds × 3 repeats, 837 ECGs). Held-out accuracy: mean of the 15 test folds with the corrected 95 % CI. Macro-F1, Mild F1, balanced accuracy and recall: from the confusion matrix pooled within each repeat, averaged over the 3 repeats. Rec. = recall of Normal, Mild, Moderate and Severe. † Selective: scored on the 48 % of patients it answers.

#	Model	Held-out accuracy [95 % CI]	Macro-F1	Mild F1	Bal. acc.	Rec. N	Rec. Mi	Rec. Mo	Rec. S	Training acc.
1	Triage †	0.696 [0.630, 0.762]	0.372	0.049	0.367	0.99	0.03	0.19	0.27	1.000
2	LGBM-FEAT	0.548 [0.512, 0.583]	0.409	0.227	0.413	0.88	0.18	0.39	0.21	1.000
3	FUSION-MLP	0.538 [0.499, 0.578]	0.438	0.255	0.440	0.79	0.24	0.42	0.31	1.000
4	ECGF-MLP	0.536 [0.506, 0.565]	0.430	0.282	0.430	0.79	0.27	0.41	0.26	1.000
5	C3	0.533 [0.496, 0.569]	0.423	0.260	0.419	0.81	0.24	0.39	0.24	0.995
6	IMG-C3	0.532 [0.492, 0.572]	0.416	0.252	0.413	0.82	0.23	0.38	0.22	1.000
7	C1	0.517 [0.477, 0.558]	0.426	0.265	0.426	0.76	0.26	0.38	0.32	1.000
8	IMG-vote	0.503 [0.463, 0.544]	0.416	0.275	0.415	0.72	0.28	0.39	0.27	1.000
9	ECGF-linear	0.491 [0.451, 0.532]	0.408	0.282	0.407	0.70	0.29	0.36	0.28	1.000
10	HDRCLIN-LR	0.486 [0.436, 0.536]	0.415	0.252	0.429	0.68	0.24	0.35	0.45	0.589
11	HDR-LR	0.483 [0.432, 0.535]	0.391	0.210	0.409	0.72	0.17	0.32	0.42	0.529
12	CLIN-LR	0.442 [0.390, 0.493]	0.376	0.240	0.382	0.60	0.25	0.36	0.32	0.720

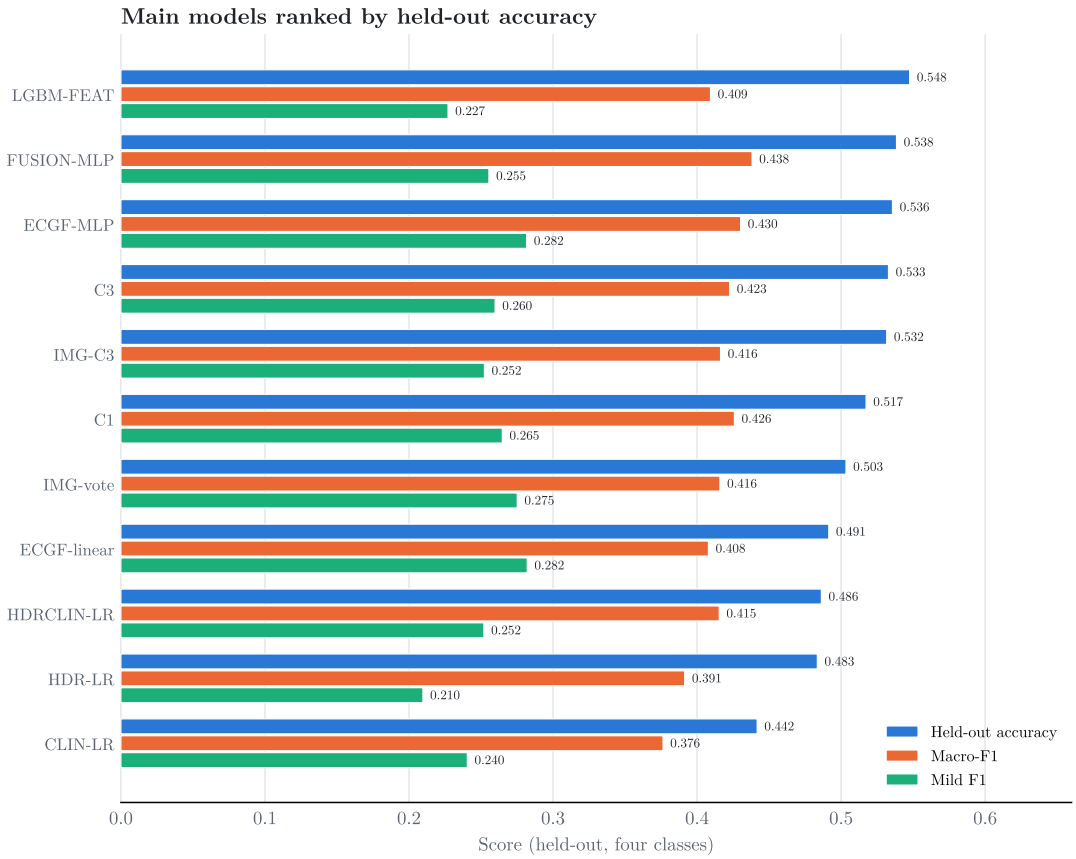


Figure 1: Held-out accuracy, macro-F1 and Mild F1 of the four-class main models (ranked by accuracy).

3 Compared with the original (early-stopped) models

Table 2: The same models before (original, early-stopped, report 45) and after training to (nearly) 100 %. Change in percentage points. Triage: original macro-F1 and Mild F1 were not recorded in its run summary.

Model	Accuracy: original → now (change)	Macro-F1: original → now (change)	Mild F1: original → now (change)
Triage	0.694 → 0.696 (+0.2)	– → 0.372	– → 0.049
LGBM-FEAT	0.524 → 0.548 (+2.4)	0.416 → 0.409 (-0.7)	0.228 → 0.227 (-0.1)
FUSION-MLP	0.544 → 0.538 (-0.6)	0.440 → 0.438 (-0.2)	0.186 → 0.255 (+6.9)
ECGF-MLP	0.542 → 0.536 (-0.6)	0.446 → 0.430 (-1.6)	0.239 → 0.282 (+4.3)
C3	0.561 → 0.533 (-2.8)	0.448 → 0.423 (-2.5)	0.221 → 0.260 (+3.9)
IMG-C3	0.558 → 0.532 (-2.6)	0.441 → 0.416 (-2.5)	0.227 → 0.252 (+2.5)
C1	0.547 → 0.517 (-3.0)	0.455 → 0.426 (-2.9)	0.255 → 0.265 (+1.0)
IMG-vote	0.541 → 0.503 (-3.8)	0.450 → 0.416 (-3.4)	0.264 → 0.275 (+1.1)
ECGF-linear	0.530 → 0.491 (-3.9)	0.444 → 0.408 (-3.6)	0.266 → 0.282 (+1.6)
HDRCLIN-LR	0.501 → 0.486 (-1.5)	0.424 → 0.415 (-0.9)	0.269 → 0.252 (-1.7)
HDR-LR	0.489 → 0.483 (-0.6)	0.393 → 0.391 (-0.2)	0.198 → 0.210 (+1.2)
CLIN-LR	0.509 → 0.442 (-6.7)	0.417 → 0.376 (-4.1)	0.205 → 0.240 (+3.5)

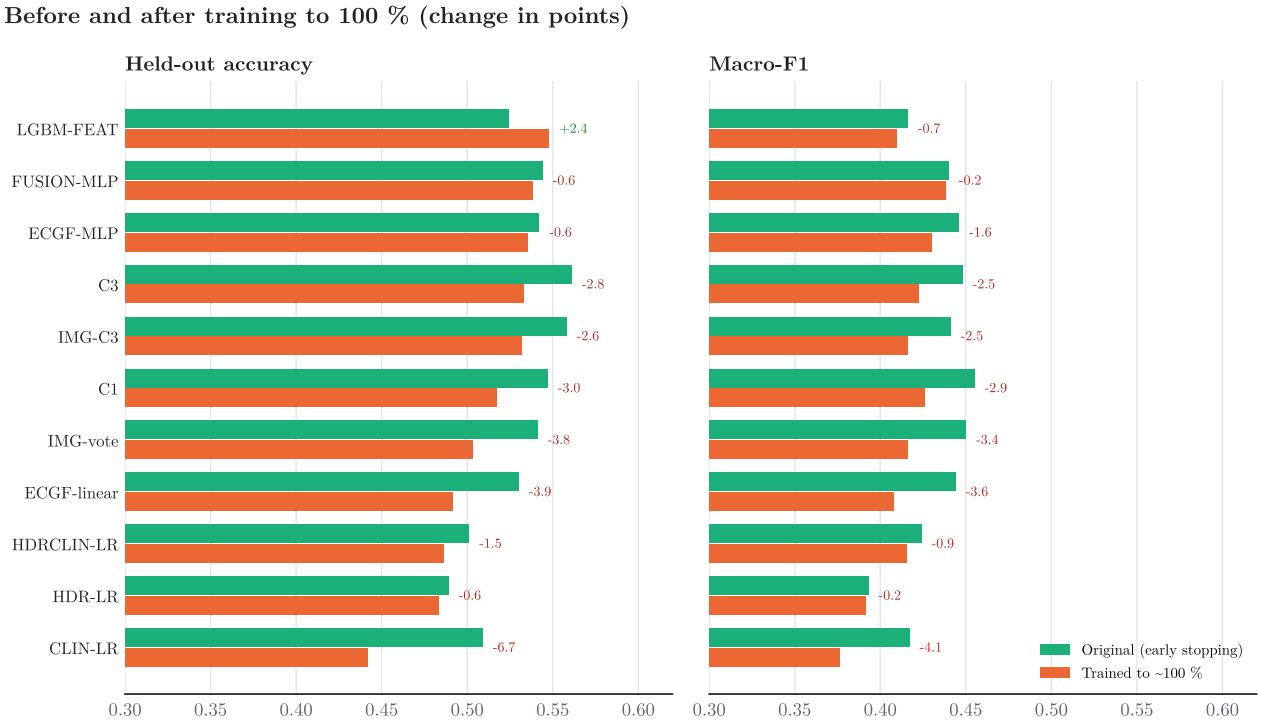


Figure 2: Original against trained-to-100 % for every four-class main model; red numbers = loss in points.

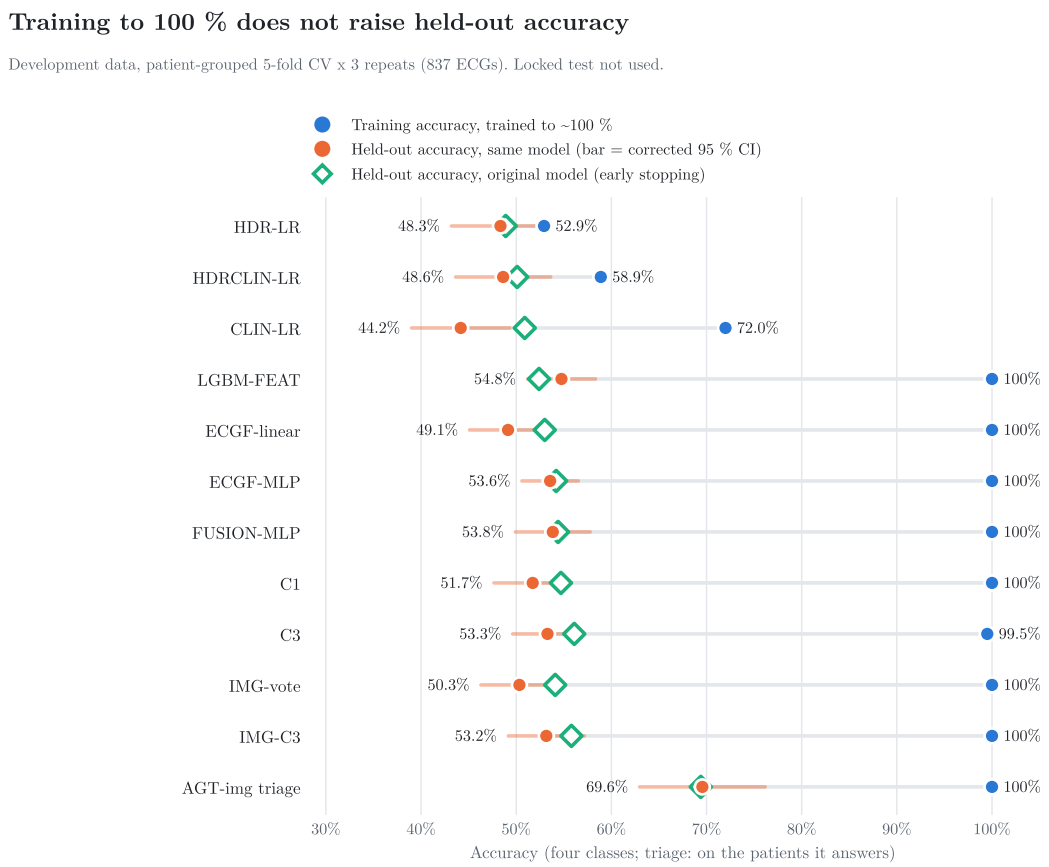


Figure 3: Training accuracy (blue) against held-out accuracy (orange, with 95 % CI) and the original held-out accuracy (green).

4 How the main models are built

Table 3: How the main models are built. ECGFounder (Net1D, 30.7 M parameters: 61 convolution and 40 linear layers in 7 stages of 20 residual blocks) is pre-trained on 10.8 million ECGs and kept frozen; only the heads are trained on the hospital ECGs. Epochs: median over the 15 folds to reach 100 % training accuracy; models without epochs show what replaces them.

Model	Inputs	Layers / structure	Parameters	Epochs
HDR-LR	16 header measurements (age, sex, rate, intervals, axes)	1 linear layer ($16 \to 4$)	68	solver (L-BFGS)
HDRCLIN-LR	16 header + 28 clinical ECG features (44 inputs)	1 linear layer ($44 \to 4$)	180	solver (L-BFGS)
CLIN-LR	118 image-only clinician features	1 linear layer ($118 \to 4$)	476	solver (L-BFGS)
LGBM-FEAT	16 header + 403 engineered waveform features (419)	2,000 boosting rounds $\times$ 4 classes, 31 leaves per tree	8,000 trees	2,000 trees
ECGF-linear	frozen ECGFounder embedding (1,024)	frozen backbone + 1 linear layer ($1{,}024 \to 4$)	4.1 k trained	100 L-BFGS iter.
ECGF-MLP	frozen ECGFounder embedding (1,024)	frozen backbone + MLP: dropout, linear $1{,}024 \to 64$, GELU, dropout, linear $64 \to 4$ (2 layers); 3 seeds	65.9 k trained	108
FUSION-MLP	16 header + ECGFounder embedding (1,040)	MLP head as ECGF-MLP (2 linear layers); 3 seeds	67.0 k trained	112
C1	FUSION-MLP + ECGF-linear + HDRCLIN-LR	average of 3 members' probabilities	members	112 (FUSION-MLP)
C3	C1	C1 + nested prior rule (one α chosen on inner OOF)	members + 1	112 (FUSION-MLP)
IMG-vote	ECGF-MLP + ECGF-linear (image only)	average of 2 members' probabilities	members	108 (ECGF-MLP)
IMG-C3	IMG-vote	IMG-vote + nested prior rule	members + 1	108 (ECGF-MLP)
Triage	ECGFounder + 118 clinician features $\to$ EF; IMG-C3	ridge + LightGBM EF regressors $\to$ EF grade (3 cut-points) + margin; answers when the grade agrees with IMG-C3	regressors + 4	4,000 trees

5 Confusion matrices

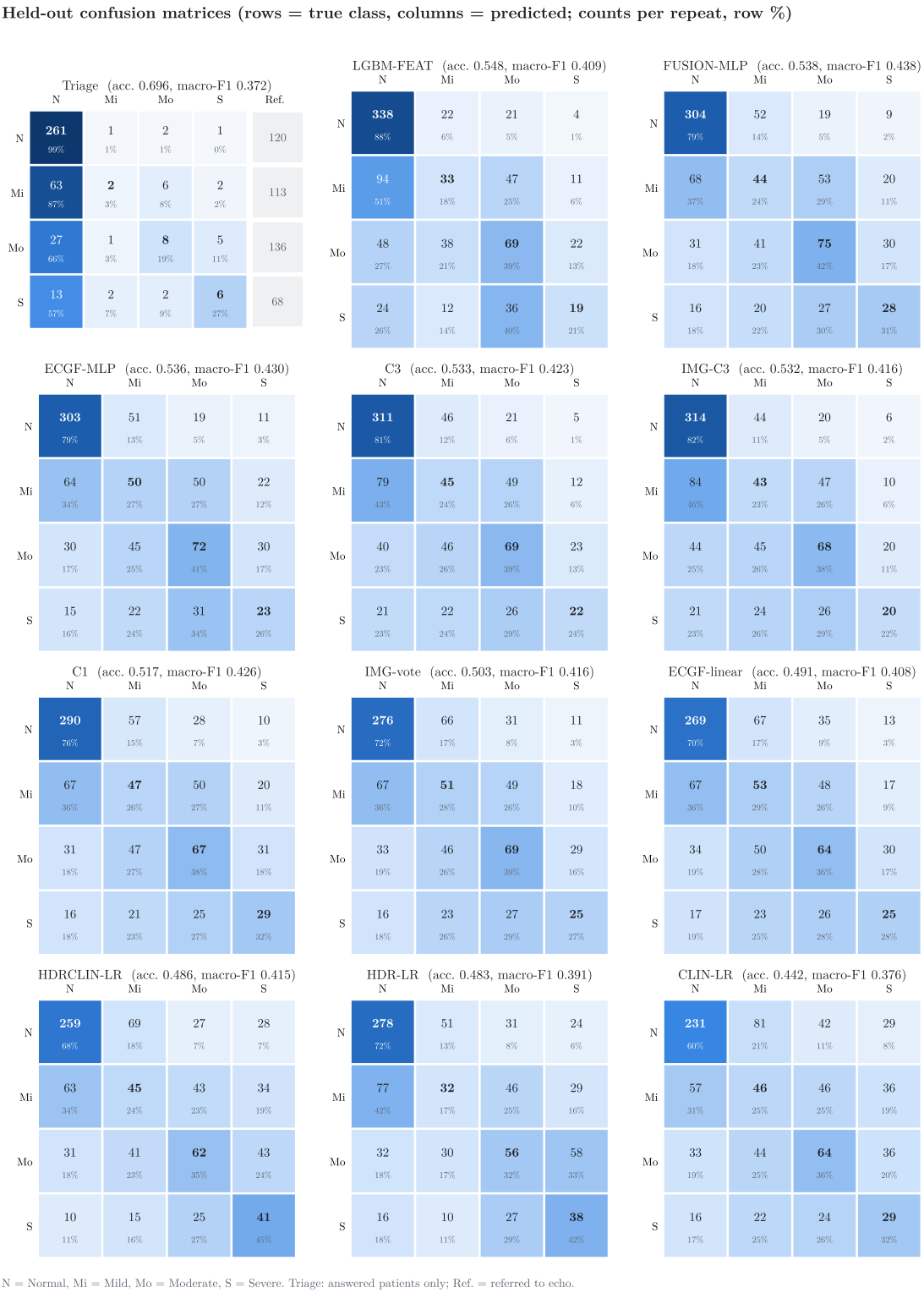


Figure 4: Held-out confusion matrices of all twelve main models, ordered by accuracy. Counts are per repeat (the 837 ECGs are predicted once per repeat); colour = row percentage (recall).



Figure 5: Recall of each class. Normal is found most often; Mild is the weakest class for every model.

6 Image and signal networks (original fixed split)

Table 4: Image and signal networks on the original fixed split (trained on 669 ECGs, validated on 168; the locked test is not loaded), ranked by validation accuracy. Best = highest validation accuracy at any epoch on the way to 100 %.

Model	Layers	Params	Epochs	Train.	Valid.	Best valid.	Orig. valid.	Change
ECGFounder fine-tuned (stages 5–6)	61 conv + 40 linear (stages 5–6 trained) + 1 FC	30.7 M	12	0.996	0.560	0.560	0.524	+3.6
XGBoost (waveform features)	2,000 trees	–	–	1.000	0.494	–	0.500	-0.6
Header 0.6 + ResNet-18 0.4	weighted average	–	–	0.972	0.458	–	0.565	-10.7
1-D SE-ResNet	37 conv + 1 FC	4.5 M	32	0.999	0.435	0.542	0.482	-4.7
ECGNet	29 conv (incl. SE) + 1 FC	6.3 M	21	0.999	0.435	0.488	0.476	-4.1
HOG + RBF-SVM	PCA-150 + RBF kernel	–	–	1.000	0.423	–	0.458	-3.5
ResNet-18	20 conv + 1 FC (18 weight layers)	11.2 M	24	0.997	0.423	0.482	0.524	-10.1
EfficientNet-B0	81 conv (16 MBConv) + 1 FC	4.0 M	37	1.000	0.417	0.512	0.351	+6.6
Logistic regression (header)	1 linear layer	–	–	0.674	0.405	–	0.500	-9.5
MobileNetV3-Large	63 conv (15 bottlenecks) + 1 FC	4.2 M	57	0.999	0.387	0.476	0.185	+20.2
ConvNeXt-Atto	40 conv (12 blocks) + 1 FC	3.4 M	53	0.996	0.345	0.518	0.452	-10.7

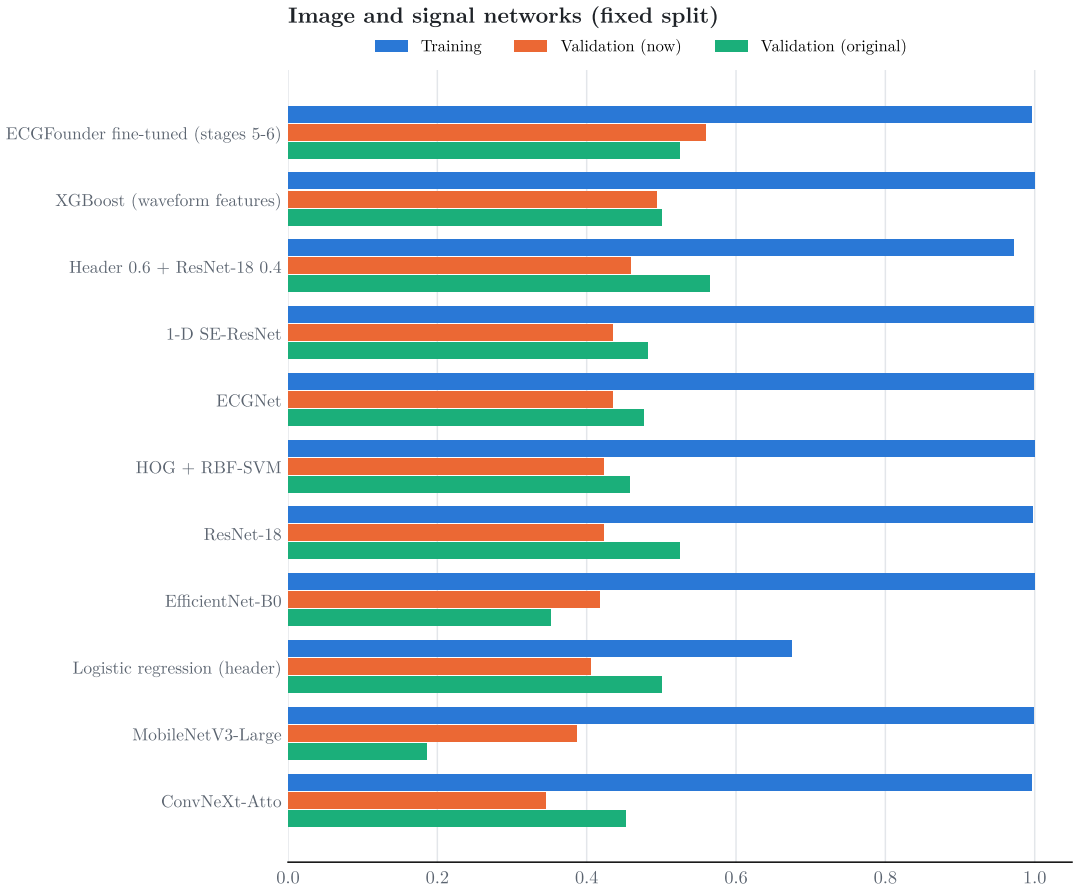


Figure 6: Training, validation and original validation accuracy of the image and signal networks.

Table 5: Variants of every phase (Phases 4, 5 and 6B, development cross-validation), ranked by held-out accuracy. SUPER-TOP10-ORD: hospital-only members; its original is the same recipe on the original members (the published 0.587 also used EchoNext).

Model	**Training acc.**	**Held-out acc. [95 % CI]**	**Macro-F1**	**Mild F1**	**Original**	**Change**
E2 LightGBM (eng + ECGF-PCA)	1.000	0.557 [0.517, 0.597]	0.422	0.208	0.544	+1.3
ENH-vote	1.000	0.549 [0.513, 0.585]	0.418	0.223	0.569	-2.0
ULT M9 kNN (k 15, distance)	1.000	0.546 [0.520, 0.571]	0.382	0.172	–	–
E6 signal-quality gating	0.988	0.542 [0.500, 0.583]	0.441	0.258	0.546	-0.4
E1 MLP, all feature families	1.000	0.542 [0.504, 0.579]	0.442	0.259	0.554	-1.2
ULT M10 gradient boosting	1.000	0.540 [0.496, 0.585]	0.409	0.223	–	–
E4 hierarchical MLP	0.999	0.540 [0.517, 0.563]	0.433	0.250	0.567	-2.7
SUPER-TOP10-ORD without EchoNext	0.783	0.539 [0.509, 0.570]	0.400	0.196	0.565	-2.6
ULT-stack (no EchoNext)	0.794	0.537 [0.500, 0.574]	0.409	0.231	0.571	-3.4
Multitask MLP (class + EF)	1.000	0.536 [0.496, 0.577]	0.434	0.258	0.546	-1.0
HIER (two-stage)	1.000	0.535 [0.488, 0.582]	0.426	0.217	0.570	-3.5
E7 mixup MLP	1.000	0.534 [0.499, 0.570]	0.434	0.263	0.552	-1.8
E3 stacking (LR, balanced)	0.826	0.532 [0.487, 0.576]	0.418	0.172	0.546	-1.4
E3 weighted average	0.985	0.530 [0.480, 0.581]	0.428	0.244	0.547	-1.7
ENH M5 LR on HIER inputs	1.000	0.501 [0.450, 0.553]	0.417	0.286	0.543	-4.2
ULT M8 LR multi-scale ECGF	1.000	0.496 [0.459, 0.532]	0.399	0.246	–	–
ENH M2 synthetic 10-s → ECGFounder	1.000	0.489 [0.439, 0.539]	0.404	0.280	0.528	-3.9
ENH M6 EF ridge → bands	0.557	0.477 [0.458, 0.496]	0.257	0.004	0.534	-5.7
ENH M4 MiniRocket	1.000	0.469 [0.423, 0.515]	0.390	0.229	0.504	-3.5
ENH M3 median-beat shape	1.000	0.412 [0.358, 0.466]	0.345	0.225	0.467	-5.5
ULT M12 zone LR	0.624	0.349 [0.302, 0.396]	0.278	0.311	–	–

7 Variants of every phase

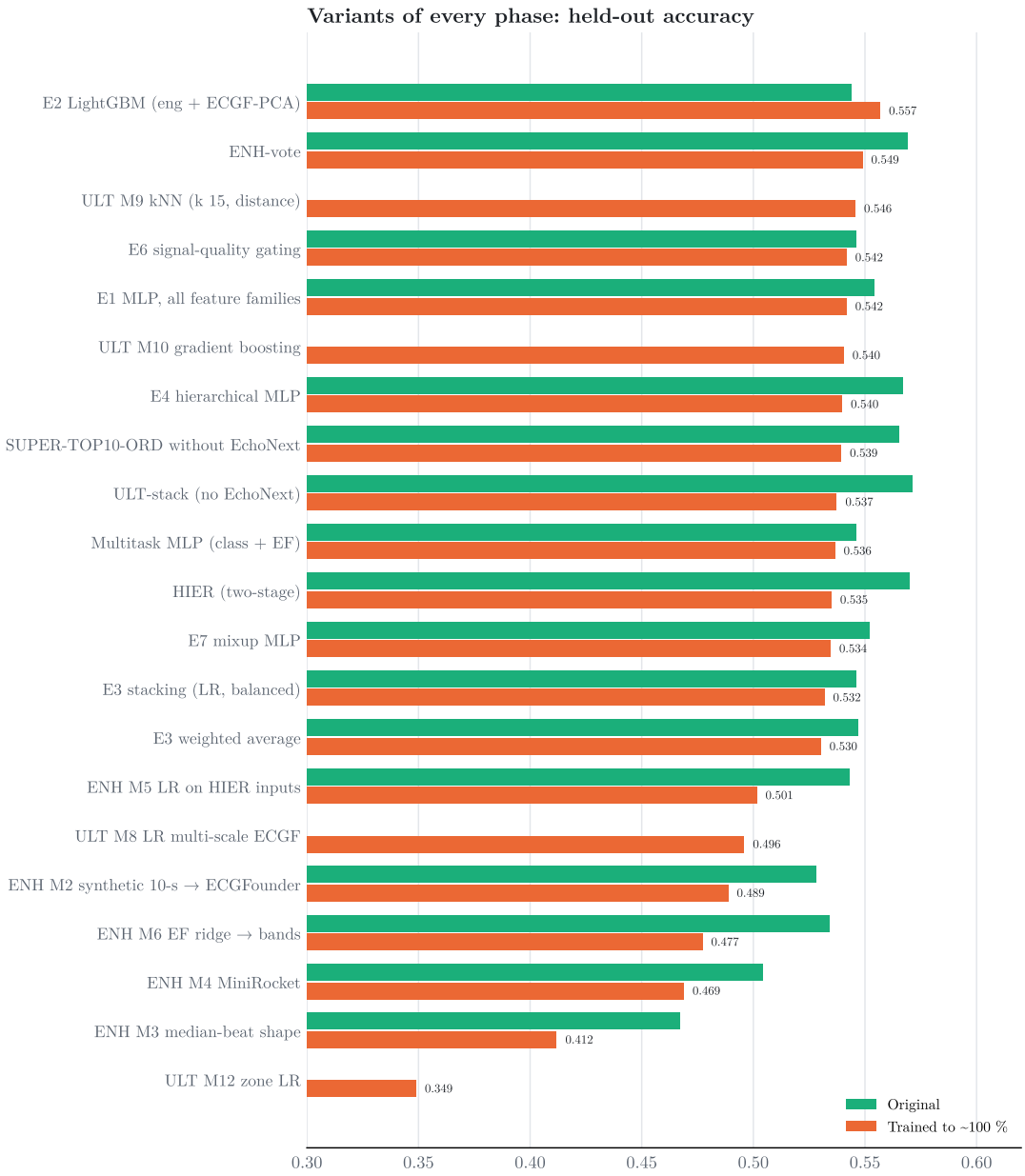


Figure 7: Variants of Phases 4, 5 and 6B: held-out accuracy of the original and of the model trained to (nearly) 100 %.

8 Label-quality studies

9 Top-3 ensembles with EchoNext (owner's laptop)

10 Training accuracy of all models

Table 6: Label-quality studies with the C3 recipe trained to (nearly) 100 %, scored on the same subsets as report Table 4.7, ranked by accuracy. Epochs: median of the FUSION-MLP member.

Study	Scored on	Epochs	Training acc.	Accuracy	Original	Change
NFS: noise-free, Severe kept	264 clean	44	0.985	0.694	0.702	-0.8
NF: noise-free	264 clean	45	1.000	0.673	0.683	-1.0
CLEAR: no border ECGs	582 clear-cut	104	0.993	0.670	0.668	+0.2
MMCLEAN: suspect Mild/Mod. dropped	620 kept	95	0.978	0.669	0.672	-0.3
HC30: reliable links only	395 reliable	52	1.000	0.583	0.594	-1.1
VER: 15 labels corrected	all 837	132	0.986	0.537	0.565	-2.8

Table 7: Top-3 ensembles retrained to (nearly) 100 % with EchoNext on the owner's laptop (numbers as reported by that session; EchoNext data never left the laptop).

Model	Training acc.	Held-out acc. [95 % CI]	Original	Change
SUPER-TOP10-ORD (best for 50s)	0.748	0.558 [0.530, 0.587]	0.587	-2.9
SUPER (greedy)	0.790	0.552 [0.507, 0.596]	0.583	-3.1
STAGED-4 (pool X)	0.722	0.548 [0.516, 0.581]	0.585	-3.7

Table 8: Members of the three ensembles (laptop run), ranked by hospital held-out accuracy. ‡ Two-class task. ° Trained on EchoNext only: training and validation accuracy on EchoNext, held-out accuracy on the hospital ECGs (H / plain input). – = not reported.

Member	Training acc.	EchoNext validation	Hospital held-out acc.
EN-MiMo-GBM (Mild vs Moderate)‡°	1.000	0.694	0.586
EN-MiMo-LR (Mild vs Moderate)‡°	1.000	0.604	0.577
H-MiMo-GBM (Mild vs Moderate)‡	1.000	–	0.569
H-TAB-GBM	1.000	–	0.548
M9	1.000	–	0.546
H-X-GBM	1.000	–	0.546
EN-gbm-X°	1.000	0.756	0.539
M10	1.000	–	0.539
ECGF-MLP	0.996	–	0.536
EN-gbm (H / plain)°	1.000	0.756	0.522 / 0.522
M5	1.000	–	0.507
PIX-ZS (CNN on GPU)°	0.999	0.725	0.505
ECGF-lin	1.000	–	0.498
EN-cls-all-X°	0.996	0.694	0.497
M8	1.000	–	0.496
PIX-ZS-band°	0.929	0.642	0.492
EN-cls-all (H / plain)°	0.999	0.673	0.490 / 0.464
EF-band	–	–	0.479
EN-cls°	1.000	0.663	0.479
EN-band°	0.781	0.520	0.479
M6 (EF bands)	0.557	–	0.477
H-TAB-LR	1.000	–	0.473
EN-band-all (H / plain)°	0.820	0.525	0.458 / 0.465
EN-mu-mid°	0.999	0.463	0.409
M12	1.000	–	0.346

Table 9: Training accuracy and epochs of every model, sorted from highest to lowest within each group.

Model	Epochs (median)	Training accuracy
(a) Main models, development cross-validation		
LightGBM on engineered features (LGBM-FEAT)	2,000 trees	1.000
ECGFounder linear head (ECGF-linear)	100 iter.	1.000
ECGFounder MLP head (ECGF-MLP)	108	1.000
Header + ECGFounder MLP fusion (FUSION-MLP)	112	1.000
C1: vote ensemble	112[b]	1.000
IMG-vote: image-only vote	108[c]	1.000
IMG-C3: image-only vote + prior rule	108[c]	1.000
Image-only triage (AGT-img)[†]	4,000 trees	1.000
MMCLEAN selective[†*]	n/r	0.999
C3: vote + nested prior rule	112[b]	0.995
Image-only clinician features, logistic regression (CLIN-LR)	solver	0.720
Header + clinical features, logistic regression (HDRCLIN-LR)	solver	0.589
Header measurements, logistic regression (HDR-LR)	solver	0.529
(b) Variants of every phase, development cross-validation		
E1 MLP, header + ECGFounder + feature families	64	1.000
E2 LightGBM, features + ECGFounder PCA	2,000 trees	1.000
E7 mixup MLP	144	1.000
Multitask MLP (class + EF)	131	1.000
HIER two-stage	108[c]	1.000
ENH M2 synthetic 10-s ECG → ECGFounder	solver	1.000
ENH M3 median-beat shape	solver	1.000
ENH M4 MiniRocket	solver	1.000
ENH M5 logistic regression	solver	1.000
ENH-vote	108[c]	1.000
ULT M9 kNN	none	1.000
ULT M10 gradient boosting	2,000 iter.	1.000
ULT M8 multi-scale ECGFounder LR	solver	1.000
E4 hierarchical MLP	52 / 39	0.999
E6 signal-quality gating	112[b]	0.988
E3 weighted average of the C1 members	112[b]	0.985
E3 stacking (logistic meta-learner)	112[b]	0.826
ULT-stack	108[c]	0.794
SUPER-TOP10-ORD, hospital members only	108[c]	0.783
ULT M12 zone LR	solver	0.624
ENH M6 EF ridge → bands	exact	0.557
(c) Label-quality studies (C3 recipe)		
HC30: reliable links only	52[b]	1.000
NF: noise-free training	45[b]	1.000
CLEAR: without border ECGs	104[b]	0.993
VER: corrected labels	132[b]	0.986
NFS: noise-free training, Severe kept	44[b]	0.985
MMCLEAN: suspect Mild or Moderate labels dropped	95[b]	0.978
(d) Image and signal networks, original fixed split		
EfficientNet-B0	37	1.000
HOG + RBF-SVM	solver	1.000
XGBoost (waveform features)	2,000 trees	1.000
MobileNetV3-Large	57	0.999

Model	Epochs (median)	Training accuracy
1-D SE-ResNet	32	0.999
ECGNet	21	0.999
ResNet-18	24	0.997
ECGFounder fine-tuned (stages 5–6)	12	0.996
ConvNeXt-Atto	53	0.996
Header 0.6 + ResNet-18 0.4	24 (ResNet-18)	0.972
Logistic regression (header)	solver	0.674
(e) Laptop run with EchoNext* (top-3 ensembles and their members)		
ECGF-lin	n/r	1.000
M5	n/r	1.000
M8	n/r	1.000
M9	n/r	1.000
M10	n/r	1.000
M12	n/r	1.000
H-TAB-LR	n/r	1.000
H-TAB-GBM	n/r	1.000
H-X-GBM	n/r	1.000
H-MiMo-GBM (Mild vs Moderate)‡	n/r	1.000
EN-cls$^{\circ}$	n/r	1.000
EN-gbm (H / plain)$^{\circ}$	n/r	1.000
EN-gbm-X$^{\circ}$	n/r	1.000
EN-MiMo-LR (Mild vs Moderate)$^{\ddagger\circ}$	n/r	1.000
EN-MiMo-GBM (Mild vs Moderate)$^{\ddagger\circ}$	n/r	1.000
EN-cls-all (H / plain)$^{\circ}$	n/r	0.999
PIX-ZS (CNN on GPU)$^{\circ}$	n/r	0.999
EN-mu-mid$^{\circ}$	n/r	0.999
ECGF-MLP	n/r	0.996
EN-cls-all-X$^{\circ}$	n/r	0.996
PIX-ZS-band$^{\circ}$	n/r	0.929
EN-band-all (H / plain)$^{\circ}$	n/r	0.820
SUPER (greedy)	n/r	0.790
EN-band$^{\circ}$	n/r	0.781
SUPER-TOP10-ORD (best for 50s)	n/r	0.748
STAGED-4 (pool X)	n/r	0.722
M6 (EF bands)	n/r	0.557
EF-band	n/r	–§

Epochs: median over the 15 folds; solver / trees / iter. = models not trained in epochs; ^{b} epochs of the neural member FUSION-MLP; ^{c} epochs of the neural member ECGF-MLP. † Selective model: training accuracy on the patients it answers. ‡ Two-class task (Mild vs Moderate). $^{\circ}$ Trained on EchoNext only (training accuracy on EchoNext). * Numbers from the owner's other / laptop session; n/r = not reported. § Not visible in the shared screenshot. Logistic regressions on few inputs cannot reach 1.000; combiners whose decision rule is tuned on cross-validated member outputs stay below 1.000 by design.

11 Where to stop training

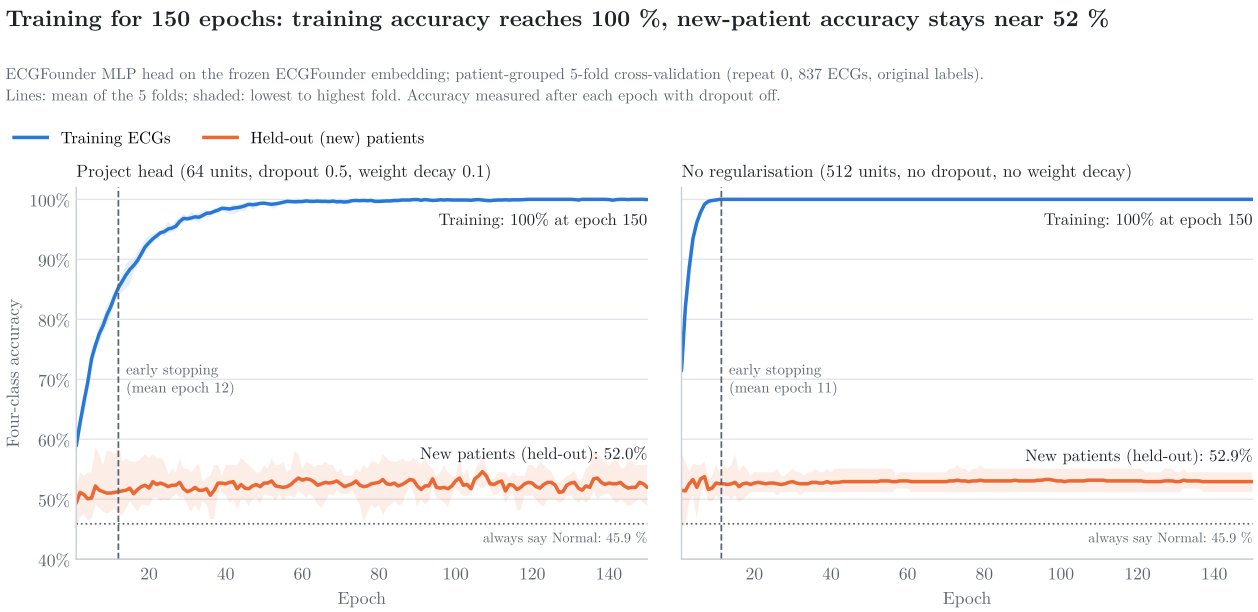


Figure 8: ECGFounder MLP head trained for 150 epochs: training accuracy reaches 100 % while held-out accuracy stays at about 0.52. The dashed line marks the early-stopping epoch.

Training to 100 % memorises the 670 training ECGs of each fold, including their label noise; that detail does not exist in new patients. The right stopping point is where accuracy on unseen data stops improving (inner-validation early stopping), at a training accuracy of about 0.60–0.85, and the held-out accuracy is the number to report.